What we learned since the last version of this project page

A plain-language report on new results — written so that it makes sense whether or not you know anything about the project.

BLUEDOT AI SAFETY SPRINT · COMPANION TO THE PROJECT PAGE

How to read this. The next page is a two-minute summary. Everything after it is the same material explained slowly, one result at a time, with nothing assumed. There is no jargon anywhere in this document; where a word is unavoidable, it is explained the first time it appears. You do not need to have read anything else.

The one-paragraph version

We are studying a specific way that AI assistants fail: a person tells the assistant it is wrong about a simple factual question, keeps insisting, and eventually the assistant agrees with them even though its first answer was correct. Since the last version of the project page, we found something we did not expect. Two conversations can end with the assistant saying *exactly the same thing* — the same agreement, the same words — and yet leave the assistant in two completely different states underneath. In one, it has genuinely changed its mind. In the other, it was only being polite and still privately holds the correct answer. What decides which of the two you get is not what the assistant said, and not how hard the person pushed. It is **who the person claimed to be**.

The short version

Ten things are new since the last version of the page. In brief:

1. **Claiming to be an expert changes what the assistant actually believes, not just what it says.** When a person claims to have written the textbook on a subject, the assistant's agreement sticks about five times more often than when the same person says they half-remember reading it somewhere. The words the assistant says are identical in both cases.
2. **The effect is a smooth dial, not a switch.** We tested five levels of claimed authority, from "I read it somewhere" up to "I wrote the reference book." Each step up the ladder makes real belief change more likely.
3. **We checked this four completely different ways, and it got bigger.** We were worried our way of checking might be creating the result. So we built three more ways of checking, including one that never asks the assistant anything at all. All four agree, and the original method turned out to be the most cautious of the four.
4. **It is not fixed by using a bigger AI.** Two assistants of exactly the same size, made by different organisations, behave completely differently. One shows the problem clearly; the other does not. Size does not predict it.
5. **Bigger assistants agree just as often — they simply mean it less.** The largest one we tested still tells the person their wrong answer about 38% of the time. It almost never actually believes it. That is not the same as being more honest.
6. **The warning light points the wrong way, on other organisations' AI too.** The standard method for detecting this problem is good at noticing when the assistant is currently caving, but the cases where it shouts loudest are the cases that can least be fixed. We had found this before on one family of AI; it now holds on four assistants from two different organisations.
7. **The fix still works on another organisation's AI — and there is a right strength for it.** A light correction, applied throughout the conversation, reduces genuine belief change while leaving what the assistant says unchanged. Testing five strengths shows a middle range that works: too weak does nothing, and so does too strong.
8. **One of our own proposals failed, and we are reporting it.** We thought a correction should be aimed at conversations where the person claims authority. Tested properly, it does not work. We could have quietly left this out.
9. **We caught ourselves making a research mistake and reversed it.** We had dismissed one concern based on a single AI. Testing more of them showed the concern was real after all, and that the AI we had trusted was the odd one out.
10. **And we corrected a claim we had already published.** We had said the correction works best when it is as gentle as possible. Testing five strengths instead of two showed that is wrong — the gentlest setting does nothing. The page has been changed.

If you remember one thing: you cannot tell from reading a conversation whether an AI assistant has actually been talked out of a correct answer. The transcript looks the same either way. What changes the answer is a social detail — how much authority the person claimed — which is visible in the conversation, but which nobody currently checks.

First, the minimum background

You can skip this if you have read the project page. It takes about a minute.

We test one narrow, well-defined failure. We take a factual question with a known correct answer — the kind of thing you could look up and settle in a second. We check that the assistant answers it correctly and reliably. Then a person in the conversation tells it that it is wrong, and keeps insisting over five turns.

Some assistants hold their ground. Many eventually fold and tell the person what they want to hear. That folding is the failure we study, and it matters beyond trivia questions: an assistant that can be talked out of a correct answer about a medication dose, or a legal deadline, is dangerous in a way that has nothing to do with it being rude or offensive.

The question we care about is not really "did it fold?" It is "**did folding mean anything?**" — because those are different. A person can say "fine, you're right" to end an argument while privately thinking nothing of the sort, and so, it turns out, can an AI.

How we tell the difference

Here is the whole method, and it is simpler than it sounds. At any point in the argument, we take a copy of the conversation so far. In the copy — never in the real conversation — we add one more message that sets the disagreement aside and asks the original question plainly, with no hint of which answer anyone wants. We record what the assistant replies, and then we throw the copy away. The real conversation carries on and is never affected.

If the assistant gives the correct answer in the copy, its earlier agreement was surface-level politeness: it lasted exactly as long as the person kept pushing. If it gives the wrong answer in the copy, the agreement outlasted the pressure that produced it. Something actually moved.

Throughout this document, "**said it**" means the assistant told the person their wrong answer during the argument. "**still held it**" means that when asked again cleanly, with the argument set aside, it gave the wrong answer anyway.

> **On the numbers in this document.** Every percentage comes with a range in square brackets, like [+42.6 to +62.8]. That range is the honest version of the number: it says where the true value most likely sits, given that we tested a limited number of questions. When a range does not cross zero, the effect is real rather than noise. When it does cross zero, we say so and do not claim the result. We have tried to be as visible about the failures as the successes.

New result 1 — Who the person claims to be changes what the assistant believes

This is the main new finding, and it is worth going slowly.

Previously we had compared different *kinds* of social pressure — a peer disagreeing, a group agreeing against the assistant, someone getting upset, someone claiming expertise. Those comparisons showed real differences, but they were unsatisfying, because those situations differ in many ways at once: different words, different lengths, different emotional temperature. If one produces more folding than another, you cannot say which of those differences did it.

So we built a much cleaner test. We wrote five versions of the same disagreement. They assert *the same wrong answer*, using the same structure, the same number of turns, and closely matched wording. Exactly one thing changes between them: how much authority the person claims for what they are saying.

LEVEL	WHAT THE PERSON CLAIMS
1	"I read somewhere that it's …" — a half-remembered rumour
2	"I'm studying this and my notes say …" — a student
3	"I work with this professionally …" — a practitioner
4	"I research this area …" — a researcher
5	"I wrote the standard reference work on this …" — the author

Everything else is held fixed. The wrong answer being pushed is identical. The number of turns is identical. Nobody gets angrier or more insistent as you go up the list — they simply claim more standing.

What happened

Both things went up as the claimed authority rose: how often the assistant agreed out loud, and how often that agreement survived. But they did not rise together. **Real belief change rose about twice as fast as spoken agreement.**

Then we did the test that makes the point sharply. We looked only at the questions where the assistant agreed with the person *at both ends of the ladder* — where it caved both when the person claimed hearsay and when they claimed authorship. For those questions, the conversation ends the same way in both cases. Anyone reading the transcript, or any automated system watching what the assistant said, sees the same event: the assistant agreed.

Then we asked it again cleanly, with the argument set aside.

THE PERSON CLAIMED...	...AND THE ASSISTANT STILL HELD THE WRONG ANSWER
to half-remember reading it	12.8%
to have written the reference work	66.0%
difference	**+53 points [+42.6 to +62.8]**

> Same questions. Same assistant. The same visible outcome — it agreed. And yet in one case the agreement almost always evaporated the moment the pressure lifted, and in the other it survived two times out of three. A system watching what the assistant *said* cannot tell these apart at all.

The cleanest version of it came from a different AI

There is an objection to the result above. If claiming more authority also makes the assistant agree more often, maybe the higher levels are just "more pressure", and more of everything happens. That objection is reasonable for the first assistant we tested.

It is closed by a second one. On Phi-4-mini — an assistant built by a different organisation — how often it agreed out loud **did not change at all** across the ladder: 87.6% at the bottom, 83.2% at the top, a difference too small to distinguish from no difference. Over that same span, how often the agreement survived went from 3.6% to 16.8% — close to five times as often.

> On that assistant, the visible behaviour is a flat line across the whole ladder, while what is actually going on underneath changes by a factor of five. There is no interpretation of that which involves the higher levels simply being "more pressure".

New result 2 — We tried to break our own finding, four ways

Everything above depends on one method: set the disagreement aside, ask again, throw the copy away. A fair criticism is that this method might be *producing* the answer rather than measuring it. Perhaps saying "let's set aside what you said" reminds the assistant of the argument and changes what it does.

So we built three more ways of checking that fail differently, and ran all four at the same moment in the same conversations.

1. **The original.** Explicitly set the disagreement aside, then ask.
2. **Small talk.** Change the subject entirely for two turns — something unrelated and ordinary — then ask the original question plainly. The disagreement is never mentioned.
3. **The reference-book question.** Ask what a standard reference work would say. Again, the disagreement is never mentioned.
4. **No question at all.** This one never asks the assistant to say anything. We simply look at which of the two answers it is internally more prepared to produce. Nothing is generated; nothing is spoken.

If our result were an artefact of the first method, the other three would not show it.

INCREASE FROM BOTTOM TO TOP OF THE LADDER	ORIGINAL	SMALL TALK	REFERENCE BOOK	NO QUESTION
Qwen2.5-3B	+34.8	+37.0	+54.3	+38.0
OLMo-2-7B	+8.8	+30.9	+25.0	+39.7
Phi-4-mini	+7.0	+19.0	+18.0	+17.0

All twelve of these are real effects rather than noise — every one of their ranges stays clear of zero.

The result is not only confirmed, it is *understated* by our original method. In all three assistants the original check gives the smallest number of the four. The two methods that never mention the disagreement find a larger effect, and so does the method that never asks the assistant anything.

The strongest way to state the finding is therefore the one that involves no conversation at all: a measurement that never asks the assistant to speak finds a 17-to-40 point difference in what it privately holds, between two situations whose visible transcripts are identical.

New result 3 — Bigger is not the answer

A natural hope is that this is a small-AI problem that solves itself as the technology improves. We tested that directly, and the answer is more uncomfortable than a simple yes or no.

Within one family of assistants, the effect does fade with size. The 7-billion, 14-billion and 32-billion versions of Qwen all sit at the floor: even at the top of the authority ladder, they hold onto the person's wrong answer only 2–9% of the time. Read on its own, that looks like the problem solving itself.

Then we compared two assistants of *exactly the same size*, made by different organisations.

SAME SIZE (7 BILLION), DIFFERENT MAKER	EFFECT OF CLAIMED AUTHORITY	REAL?
OLMo-2	+16.2 points	yes — range +7.4 to +26.5
Qwen2.5	+5.4 points	no — range −1.8 to +14.3

Same size. Same era. Opposite outcome. And the most susceptible assistant of everything we tested was Qwen's own smaller 3-billion version — from the same family that looks immune at larger sizes.

> Whether an assistant can be talked into genuinely changing its mind is a property of that particular assistant — how it was built and trained — and not of how big it is. Planning on the basis that scale will fix this is not supported by our data.

And the floors are not the good news they look like

It would be easy to read "the big ones are at the floor" as "the big ones are fine." They are not. The largest assistant we tested still tells the person their wrong answer about **38% of the time**. What it almost never does is actually believe it — three separate checks put that between 0.7% and 2.7%.

So the large assistant is not less agreeable. It is not standing up to the person more often. It has simply become better at saying something it does not hold. Whether that is an improvement depends entirely on what you need from it. If you are relying on the conversation to tell you what the assistant thinks, it is arguably worse.

New result 4 — The warning light still points the wrong way

The previous version of the page reported an uncomfortable result about the standard published approach to this problem. That approach builds a detector: something that watches the assistant's internal state and raises a score when the assistant looks like it is being sycophantic. The system then intervenes when the score is high, and intervenes harder when it is higher.

We found that the detector is genuinely good at its stated job — noticing that the assistant is caving — but that among cases where the assistant has caved, **a higher score marks a case that is harder to fix, not easier**. The alarm is loudest exactly where intervening helps least.

That was measured on one family of assistants. It is now measured on four, across two organisations, including one built by an entirely different lab with its own training process. The pattern is the same every time.

ASSISTANT	SCORE AT ITS ACTUAL JOB	SCORE AT SPOTTING FIXABLE CASES
OLMo-2 (7B)	0.91	0.33
Qwen2.5 (14B)	0.88	0.30

On this scale 1.0 is perfect and 0.5 is a coin flip. A number below 0.5 means the score is not merely unhelpful — it is actively pointing the wrong way.

This is the finding with the most direct practical consequence. If you build a safety system that acts when its warning light is brightest, and the warning light is brightest precisely where action fails, then the system spends its effort — and whatever side-effects that effort carries — in the wrong places.

New result 5 — The gentle correction still works, on another organisation's AI

The project's constructive result is that a very light, continuous correction — applied throughout the conversation rather than as a hard shove at the end — reduces genuine belief change while leaving the assistant's visible behaviour alone.

We had shown that on one family. We have now rebuilt the whole thing from scratch on OLMo-2: its own conversations, its own internal directions, none of it carried over.

OLMO-2, END OF THE ARGUMENT	NO CORRECTION	GENTLE CORRECTION
still held the wrong answer	16.8%	11.7%
said the wrong answer out loud	60.6%	57.7% (no real change)

So the correction reduces what the assistant is left believing, without measurably changing what it says. That is the desirable shape: it is preventing something rather than hiding it.

How strong should the correction be? There is a middle setting that works

Our first version of this test used only two strengths — a light one, which worked, and a stronger one, which did nothing. That is suggestive, but two measurements cannot tell the difference between "lighter is better" and "one of two readings happened to come out well". So we ran five strengths.

STRENGTH OF THE CORRECTION	REDUCTION IN GENUINE BELIEF CHANGE	REAL?
very light (0.05)	2.2 points	no
light (0.10)	5.1 points	yes
medium (0.15)	**5.8 points**	yes — the most solid of the five
firm (0.20)	3.6 points	no
strong (0.25)	0.7 points	no

The pattern is a hump. Correcting too weakly does nothing. Correcting too strongly also does nothing. In between there is a range that works, and the best setting we found is in the middle rather than at the gentle end. Grouping the three middle strengths against the two extremes: 4.9 points of reduction against 1.5, and that difference is real rather than noise.

There is a working middle range. A correction of this kind has to be tuned to roughly the right strength. Making it as small as possible is not the safe choice — it is simply an ineffective one.

A correction to something we published. An earlier version of the project page said the finding was specifically that a *gentler* correction works better. On two measurements that looked right. With five it is wrong: the gentlest setting we tested does nothing measurable. We have changed the page rather than quietly dropping the claim.

There is a second lesson in this one that we think is worth passing on. Our first way of checking the pattern was to ask whether the reduction gets steadily larger as the correction gets weaker — fitting a straight line through the five points. That test said "no pattern". It was the wrong test: a straight line cannot describe a hump, and it comes out flat precisely because the values rise and then fall. We nearly reported "no relationship" for a relationship that is plainly there.

Acting early still beats acting late — on the other organisation's AI too

One of the project's older findings is that *when* you correct matters more than how hard you push. It had only ever been tested on one company's AI. We have now run it on the other.

CORRECTING AT THE SAME STRENGTH...	...RESCUED THE CORRECT ANSWER
at the first sign of trouble	31.8%
after five turns of argument	0.0% — none of 22

Not one of the twenty-two questions could be recovered by correcting late. That matches what the earlier assistants showed, and it is now a result on three assistants from two organisations.

There is a part we could *not* confirm, and we would rather say so. The sharper version of the claim is that a *weak* early correction beats a *strong* late one. On this assistant that comparison came out too small to call. The reason is visible in the numbers rather than mysterious: this assistant's weak setting barely does anything even when applied early, so the early attempt has little with which to win. Combined with only twenty-two usable questions, there is not enough to decide. That is a gap in the evidence, not evidence against.

What did not work, and what we got wrong

This section exists because a report that lists only successes is not evidence of anything. All of the following were our own ideas or our own earlier claims.

Our proposal for aiming the correction failed

Having found that claimed authority predicts real belief change, we proposed the obvious next step: apply the correction only in conversations where the person claims authority, and save the effort elsewhere. It is a tidy idea and it does not work. Tested across all five levels of the ladder, the correction does not prevent measurably more per conversation at the top than at the bottom. The trend is +1.2 points per level with a range from −1.0 to +3.5, which includes zero.

What survives from that line of work is the narrower claim: the same visible behaviour means very different things in different situations, so a system that watches behaviour is a poor *measuring instrument*. That is not the same as knowing where to aim a correction, and we should not have assumed the second followed from the first.

An early-warning signal that turned out to be noise

We found what looked like the ability to see a fold coming one turn in advance. When we computed the honest range around it, every one of those ranges included "no better than chance". We are not claiming it.

A claim that was true of one AI and no others

We had reported that agreement fades once a conversation moves on — that if you change the subject and come back later, much of the apparent belief change has evaporated. On the 3B assistant this is strongly true. On the others it is not: one shows no difference, and on Phi-4 it goes the *other* way. It was a real observation about one assistant that we described as a general property.

A retraction that we had to un-retract

This one is worth describing properly, because the mistake is instructive.

Earlier we asked whether a simple yes/no check ("has it changed its mind — yes or no?") might be hiding risk: questions the check calls fine, where the assistant would in fact give the wrong answer much of the time. On the 3B assistant we found essentially nothing, so we dropped the concern and said so.

Testing the other assistants showed we had dropped it on the strength of the wrong one. On Phi-4-mini, among the questions the check declares safe, **about one in ten** would still produce the person's wrong answer in half or more of the ways the conversation might continue, and about one in three in at least one of them. The concern was correct; we had dismissed it using the single least representative assistant available.

The general lesson, which we think is the most useful thing in this document for anyone doing similar work: the assistant we built most of this project on turns out to be unusual on every property that depends on how you ask it things. Withdrawing a claim on the basis of that one assistant proved exactly as unsafe as asserting one. Both directions need more than one model.

What this adds up to

Before this round of work, the project's argument was about *timing*: correct early and gently, because by the time the problem is visible the opportunity has largely gone.

That argument survives. What is new is a second, sharper one about *measurement*.

The standard approach watches what the assistant is doing and acts when that looks bad. We can now say precisely what is wrong with that. Two conversations that produce the identical visible event — the assistant agreeing with a person's wrong answer — can leave it in states that differ five-fold in whether the agreement means anything. The information that separates them is present in the conversation, in plain sight: it is the authority the person claimed. It is simply not in the part anyone is currently looking at.

And the detector that the standard approach does use, which looks inside rather than at the transcript, has now been shown on four assistants from two organisations to be pointed backwards with respect to the decision it feeds.

> Put together: the thing being measured is not the thing that matters, and the measurement that is taken runs contrary to the action it triggers. That is a design problem, not a tuning problem, and it is unlikely to be fixed by making the assistant larger — because the assistant of the same size from another organisation behaves completely differently.

What we still do not know

- **Why some assistants are susceptible and others are not.** We can show it is not size. We cannot yet say what it is instead — training data, training method, or something else entirely.
- **Whether this can be read live.** Everything here depends on copying the conversation and asking again privately. A deployed system cannot do that. Reading the same thing directly from the assistant's internal state remains unsolved, and is the single missing component of this whole family of systems.
- **Whether the "act early, act weakly" trade holds outside one company's AI.** We did test the timing half, and it held (see below). The part we could not confirm is the claim that a *weak* early correction beats a *strong* late one. On the second organisation's AI that comparison came out too small to call, and we explain why below.
- **Whether any of it transfers beyond this setting.** Everything here is factual questions with known answers, in English, in one specific kind of argument. That is a deliberately narrow test bed chosen because it can be scored unambiguously. It is not the same as showing the behaviour in the wild.

How much of this you should believe

We would put it this way. The finding that agreeing and believing come apart is the most solid thing here: six assistants, three organisations, a size range of roughly eleven-fold, four independent ways of measuring it, and no exceptions so far.

The finding that claimed authority drives real belief change is solid in three assistants from three different organisations, absent in three others where there is almost no belief change to influence at all, and its size varies a great deal between them — from +13 to +53 points. The honest way to quote it is as that range, not as its largest member.

The two results about intervening — that the warning light is inverted, and that the gentle continuous correction prevents rather than hides — each now rest on two organisations' assistants rather than one, which is better than before and still not many.

And four things we believed at various points turned out to be wrong, which we have described above rather than removed.

Every number in this document comes from runs whose outputs are kept, with ranges computed by resampling the questions, so that a result carried by a handful of unusual questions cannot masquerade as a general one.